

UNIVERSITY OF TECHNOLOGY, JAMAICA
SCHOOL OF COMPUTING & INFORMATION TECHNOLOGY
Programming 2: Tutorial/Practical Worksheet #5

Tutorial Questions

1. Define an array in your own words.
2. If an array starts at memory address 1000 and each integer takes 4 bytes, what is the address of arr[3]?
3. What is the range of index values of the following array? int arr[100];
4. What is the size of the following array? int arr[] = {1,1,1,1,1,1};
5. What is the value assigned to arr[2] in the following array initialization? int arr[3] = {3,2,1}
6. Given an array A[6] = {4,5,2,9,3,7}. What is the value of A[5] + A[1] * A[2]?
7. Consider the following 2D array in C: int grid[3][3] = {{1, 2, 3}, {4, 5, 6}, {7, 8, 9}}; Calculate the result of: grid[0][0] + grid[2][2]) * grid[1][1] – grid[1][2]
8. Given an array A[5][6] = {{1, 2, 3, 4, 5}, {2, 4, 6, 8, 10}, {20, 10, 5, 3, 1}, {3, 6, 9, 12, 15}}. Write the pseudocode to fill in this row and column with the totals of each column, each row, and the grand total.

1	2	3	4	5	
2	4	6	8	10	
20	10	5	3	1	
3	6	9	12	15	

In the Laboratory

Complete any unfinished questions from the tutorial section. Implement and execute the following programs using both **C** and **Python**:

1. Write a program that will allow the user to populate a 10-element array with integer values. The user should then be asked for a value to be retrieved from the array. The program should then search for the value and display the index of the array the value was found. If the value is not found/stored in the array an appropriate message should be displayed.

2. Create a program that will populate an array of size 10 with random numbers between 10 and 25. Pass this array to a function that will determine and display the smallest number and a second function that will determine and display the largest number in the array.
3. A bank wants to simulate balances for 25 customers. Generate balances between \$1,000 and \$50,000 and store in a 1D array. Create functions to:
 - a. Count customers with balance above \$25,000.
 - b. Calculate total money in the bank.
 - c. Find the lowest balance.
 - d. Advanced: Create a function that adds 5% interest to each account.
4. A hospital monitors 10 patients over 7 days.
 - a. Rows → Patients
 - b. Columns → Days
 - c. Store daily heart rate readings (random between 60–120 bpm).

Create functions to:

- a. Find the average heart rate per patient.
- b. Identify patients with any reading above 100.
- c. Find the overall highest heart rate recorded.
- d. Return both patient number and day of highest reading.

Challenge

A retail store records the number of items sold in 30 transactions during the day. Due to repeated purchases of the same quantity, some values may appear more than once. You are required to write a program to determine how many duplicate elements exist in the array.